9/10/2026

Weak law of large numbers

independent, identically distributed

$$X_1, X_2, \dots, X_n \sim X$$

$$\lim_{n\to\infty} \frac{\sum_{i=1}^{n} x_i}{n} \longrightarrow E[X] \text{ with prob} = 1$$

Random walks in graph



$$P_0 = \left(P_0^{\,1}, P_0^{\,2}, P_0^{\,3}\right)$$

$p_0 = (1, 0, 0)$

$p_1 = (0, \frac{1}{2}, \frac{1}{2})$ $\quad p_2 = (\frac{1}{2}, \frac{1}{4}, \frac{1}{4})$

$p_{t+1} = p_t P$

$p_t = p_0 P^t$

$$P = \begin{pmatrix} 0 & \frac{1}{2} & \frac{1}{2} \\ \frac{1}{2} & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{pmatrix}$$

L_1 norm of rows = 1 → row stochastic matrix

$p_1 = p_0 P$

$p_2 = p_1 P = p_0 P^2$

Show that p_{t+1} is a probability vector, assume p_t is.

$p_t^i \rightarrow$ probability token will be at vertex i at time t

$$P_{uv} = Pr(u \rightarrow v)$$

$$= Pr(x_{t+1} = v \mid x_t = u)$$

$$P = \frac{A}{d} \quad \leftarrow \text{adjacency matrix}$$

$\leftarrow$ degree

$$P = D^{-1}A = \begin{pmatrix} d_1 & & \\ & \ddots & \\ & & d_n \end{pmatrix} A$$

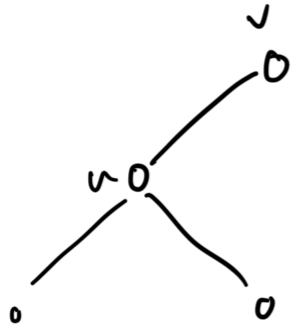
As $t \rightarrow \infty$, $p_t \rightarrow \pi$ (Stationary distribution)

Conditions for stationary.

(1) graph should be connected

(2) non-bipartite

$$\pi P = \pi \cdot 1$$



$$(\pi P)_u = \sum_{v \in \text{neg}(u)} \pi_v$$